

Rongfeng Cheng

Medical AI Researcher | Trustworthy ECG & Multimodal Learning

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RESEARCH PROFILE

Undergraduate researcher at South China University of Technology studying whether medical AI models use clinically relevant evidence, how their behavior changes across datasets, and how physiological signals can be aligned with language. Two first-author manuscripts are under review at IEEE JBHI and IEEE BIBM 2026.

EDUCATION

South China University of Technology (SCUT)

B.S. in Big Data Management and Application | 2023 - 2027

- GPA: 3.42/4.0; ranking: 80/156; average: 83.4/100.
- Selected coursework: Deep Learning, Data Structures and Algorithms, Optimization, Mathematical Modeling, Big Data Systems, and Data Governance.

RESEARCH EXPERIENCE

Evidence-Use Auditing for Deep ECG Models

Research Lead, First Author | Jan 21, 2026 - Present

- Developed a matched-control perturbation framework to test whether deep ECG classifiers depend on clinically relevant evidence rather than task-correlated shortcuts.
- Evaluated PTB-XL and CODE-15% across architectures, perturbation doses, clinical evidence windows, and frozen-weight cross-dataset transfer.
- Manuscript under review at IEEE Journal of Biomedical and Health Informatics (JBHI).

DLEF: Dataset-Label Evidence Fingerprints

Research Lead, First Author | Jun 28, 2026 - Present

- Proposed directional and boundary-aware evidence fingerprints to characterize dataset-label evidence structure.
- Studied cross-dataset AUROC degradation across CODE-15%, PTB-XL, and Chapman-Shaoxing.
- Manuscript under review at IEEE BIBM 2026.

PathCLIP: Knowledge-Aware ECG-Text Representation Learning

Research Lead | Jul 25, 2026 - Present

- Investigating diagnostic hierarchies, ECG findings, and evidence grounding for ECG-text alignment and zero-shot recognition.
- Designed matched-count, permuted-hierarchy, residual-only, and held-out-label controls; current work shifts from ontology-only supervision toward finding-level evidence alignment.

PUBLICATIONS

Auditing Knowledge-Aligned Evidence Use in Deep Learning Models: A Matched-Control Study in ECG Classification

First author. Under review at IEEE JBHI.

DLEF: Dataset-Label Evidence Fingerprints for Cross-Dataset ECG Generalization

First author. Under review at IEEE BIBM 2026.

CodeSafetyBench: Evaluating and Improving Ethical Safety in Large Language Model Code Generation

Co-author. iScience, 2026. DOI: 10.1016/j.isci.2026.115073.

Artificial Intelligence Applications in X-ray Coronary Angiography: A Comprehensive Review of Image Analysis Techniques

Co-author. HNNDL 2026. DOI: 10.1145/3795892.3795917.

Artificial Intelligence for Echocardiographic Assessment of Pulmonary Hypertension and Right Heart Function: Advances in Intelligent Diagnosis and Risk Stratification

Co-author. HNNDL 2026. DOI: 10.1145/3795892.3795918.

PROFESSIONAL EXPERIENCE

- AI Research Institute, Guangdong No. 2 Provincial People's Hospital**
Research Intern | Nov 2024 - Present
 - Conduct research on medical LLMs, cardiovascular knowledge representation, clinical question answering, and AI product prototyping.
 - Contributed to a standardized-patient Q&A; pipeline built from about 8,000 clinical cases, a myocardial-ischemia knowledge graph, patents, software copyrights, and clinical AI prototypes.
- Yunzhan Zhichuang Technology Co.**
Tech Intern | 2025 - Present
 - Translate AI health research outputs and prototypes into user-facing applications.

SELECTED PROJECTS

- KG-Based Standardized Patient Q&A;**
Medical LLM / Knowledge Representation
 - Processed approximately 8,000 cardiovascular case PDFs; built an LLM-based extraction pipeline with NER, relation extraction, entity disambiguation, Neo4j storage, and hybrid RAG.
- Myocardial Ischemia Knowledge Graph**
Project Lead
 - Built a domain knowledge graph with collaborative ontology management, XGBoost-based implicit reasoning, and graph-mapped interpretability analysis.
- IHD Trajectory Similarity Retrieval**
Clinical AI / MIMIC-IV
 - Developed trajectory retrieval and post-cutoff validation for ischemic heart disease ICU stays using time-windowed clinical variables and precomputed k-NN similarity graphs.

PATENTS AND INTELLECTUAL PROPERTY

- Method, System, and Electronic Device for Medical Question-Answering Output Assurance - First Inventor; application #202610582199.6; filed Apr 2026; preliminary examination passed.
- Knowledge Graph Ontology Intelligent Management System V1.0 - Software Copyright #2025SR2454864.
- LLM-Based Cardiovascular Proactive Health Management Intelligent System V1.0 - Software Copyright #2025SR2080673.
- Two additional patent applications in progress.

HONORS

- First Prize, 2025 Challenge Cup (Jinan University Campus).
- Second Prize, 2025 9th Guangzhou Mathematical Modeling University League.
- Third Prize, 2025 National Mathematical Modeling Contest - Guangdong Division.
- National Innovation and Entrepreneurship Training Program - Project Completed.

SKILLS

Programming Python, TypeScript, SQL	AI / ML PyTorch, XGBoost, ECG signal processing, representation learning
Medical AI Evidence-use auditing, multimodal learning, LLM evaluation, RAG	Web / Data React, Next.js, Flask, Neo4j, Git, Docker

LANGUAGES

Chinese (native) | Cantonese (native) | English: IELTS 6.5, CET-6 528